FL:- $\frac{\Sigma \text{membership} \times \text{corresponding o/p}}{\Sigma \text{membership}}$

$= \frac{0.2 \times 20 + 0.5 \times 40 + 0.3 \times 60}{0.2 + 0.5 + 0.3}$

$= \frac{42}{1}$

$= 42$

⇒ An automatic Airconditioner uses fuzzy to calculate cooling output based on room temperature

temp	membership value	cooling o/p
low	0.2	200
med	0.3	500
high	0.5	900

⇒ FL:- $\frac{\Sigma \text{membership} \times \text{corresponding o/p}}{\Sigma \text{membership}}$

$= \frac{0.2 \times 200 + 0.3 \times 500 + 0.5 \times 900}{0.2 + 0.3 + 0.5}$

$= \frac{640}{1}$

$= 640$ ₹

applications

→ Home ~~applications~~ appliances
→ auto mobile systems
→ medical systems
→ robotics
→ trapic control
→ decision making systems
→ whether for

Machine learning

It is a branch of AI that enables computer to learn from data and make predictions or decisions without being program for every task.

Data

Raw fact value used to train machine learning model.

features

Ex:-

RH	study hours	P/F → features
1	2	F
1	3	F
3	4	P
4	5	P
3	6	P

→ label

features

They are the input variables used by machine learning modle to make prediction.

label

It is the target output for the answer that a ML modle learns

model

It is a learned representation of ~~patterns~~ patrans in data used to make prediction or decisions

training

It is proces of teaching a machine learning model using data so that it can learn patrans & relationship.

testing

It is a process of evaluating trained model, using unseeing data

validation

~~It~~ is a proces of evaluating model during development to select suitable setings

two types.
1) Supervised learning
2) Unsupervised learning

Supervised learning

It is the model which learns from labeled data.
→ A data contain both input and correct outputs.

Ex:- spam detection in emiles, student result detection, decise prediction

Unsupervised learning

The model is trained with unlabeled data
→ A data contain only input only.

Ex:- calculator,

Supervised

classification

It is a supervised learning technique used to predict or class of a given data. based on it's feature.

Ex:- emile spam detection, student result detection

steps in classification

step1: collect the data
step2: label the data
step3: train the model.
step4: predict the class.

types of classification
1) Binary classification -> emile spam detection
2) multiclass classification -> fruit classification.

multiclass
It classifice data into three or more class.

Binary class
It classifice data into only two ~~typ~~ class

Algorithem of classification.

① decision tree
② KNN
③ Naive Bayes
④ SVM

KNN (K nearest neighbours)
It is a supervised ML algorithem used in classification.
→ It classifies the data base on K value.

Note.
It is also called lazy learning algorithem because It does not belde explicit model training it store the data during training model during it uses making prediction.

steps

1. choose the k value (number of nearest neighbours (odd only)
2. calculate the distance $D = |A_{new} - A|$
3. sort the distance in acending order.
4. select k neighbours
5. predict.

Ex:- classify a student as passed or fail using KNN with k=3.

student	study hours	class
A	1	fail
B	2	fail
C	3	pass
D	4	pass
E	5	pass

⇒ A new student studies for 2.5 hours predict the class

student	distance	
A	(2.5-1) = 1.5	F
B	(2.5-2) = 0.5	F
C	(2.5-3) = 0.5	P
D	(2.5-4) = 1.5	P
E	(2.5-5) = 2.5	P

$d = |x_{new} - x|$

student	distance	rear value		
B	0.5	1	fail	f
C	0.5	1	fail	p
A	1.5	2	fail	f
D	1.5	2	fail	p
E	2.5	3	pass	p

3

K=3

A new student also fail.

→ using KNN algorithem K=3 classifie new fruit based on weight

fruit	weight	class
A	120	apple
B	130	orange
C	140	apple
D	150	orange
E	160	orange

A new fruit weight 145 predet its class

→

fruit	distance	
A	|145-120| = 25	a
B	|145-130| = 15	o
C	|145-140| = 5	a
D	|145-150| = 5	o
E	|145-160| = 15	o

fruit	distance	rear		
C	5	120	apple	a
D	5	120	apple	o
B	15	120	apple	o
E	15	120	apple	o
A	25	120	apple	a

3 → o

K=3

A new fruit is orange.

→ Using KNN algorithem K=3 classifie new person healthy or unhealthy. base on weight

person	weight	class
A	40	H
B	45	H
C	50	U
D	55	U
E	60	H

new person weight 52 kg

person	distance	
A	|52-40| = 12	H
B	|52-45| = 7	H
C	|52-50| = 2	U
D	|52-55| = 3	U
E	|52-60| = 8	H

person	distance	
C	2	U
D	3	U
B	7	H
E	8	H
A	12	H

3

new person was unhealthy

Regression

It is a supervised machine learning technique used to predict a continous numerical values based on input data.

Types of regression

Relationship b/w

→ simple Linear R → One independent variable and depdent variable

→ multiple Linear R → Relationship b/w two or more independent & depdent

→ polynomial R → It is used model a curve relation

Simple Linear Regression

$y = a + bx$

$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$ $\quad a = \frac{\Sigma y - b\Sigma x}{n}$

SName	Hours studies (x)	marks obtain(y)
A	40m	20
B	20m	40
C	30m	50
D	40m	40

if student study 50 ? How much marks obtain

SName	HS (x)	MO(y)	x^2	xy.
A	40	20	100	200
B	20	40	400	800
C	30	50	900	1500
D	40	40	1600	1600
	$\Sigma x=100$	$\Sigma y=150$	$\Sigma x^2=3000$	$\Sigma xy=4100$

$(\Sigma x)^2 = 10,000$

$n=4.$

$$b = \frac{4 \times 4100 - 100 \times 150}{4 \times 8000 - 10000} = \frac{16400 - 15000}{12000 - 10000}$$

$$= \frac{1400}{2000} = 0.7$$

$$a = \frac{150 - 0.7 \times 100}{4}$$

$$= \frac{150-70}{4} = \frac{80}{4} = 20$$

$y = a + bx$

$= 20 + 0.7 \times 50$

$y = 55$

A shop owner records amount spend on advertising and corresponding sealse values

moth	acx	sales y
1	10	300
2	20	400
3	30	600
4	40	800

if ac = 50

Month	ac (x)	sales (y)	x^2	xy.
1	10	300	100	3000
2	20	400	400	8000
3	30	600	900	18000
4	40	800	1600	32000
	$\Sigma x = 100$	$\Sigma y = 2100$	$\Sigma x^2 = 3000$	$\Sigma xy = 61000$

$(\Sigma x)^2 = 10000$

$n=4, x=50$

$$b = \frac{4 \times 61000 - 100 \times 2100}{4 \times 3000 - 10000} = \frac{244000 - 210000}{12000 - 10000}$$

$$= \frac{34000}{2000} = 17$$

$$a = \frac{2100 - 17 \times 100}{4}$$

$$= \frac{2100 - 1700}{4}$$

$= 100$

$y = a + bx$

$= 100 + 17 \times 50$

$= 100 + 850$

$= 950$

Unsupervised learning

It is a type of ML in which computer learn pattern and relationship from unlabled data with being given the correct output

Types

1) clustring
2) association

① clustring

grouping the similer data items together.

② association

Finds relationship btw items.

clustring

Kmeans

It is unsupervised ml algorithem used to divide data set into k clusters based on similer data points.

steps in k means

1) choose the number of clusters K
2) select the K initial centroids
3) calculate the distance btw each data point & centroids
4) assign each data point to its nearest centroids
5) recalculate the centroid of each cluster
6) repeat the 3 to 5 until there is no changing cluster

Ex apply kmeans algorithem to group the following data int K=2 clusters data points = 2, 4, 10, 12
initial centroids C1 = 2, C2 = 10

→ K = 2
C1 = 2, C2 = 10

Data points	C1 distance	C2 distance	cluster
2	|2-2| = 0	|2-10| = 8	C1
4	|4-2| = 2	|4-10| = 6	C1
10	|10-2| = 8	|10-10| = 0	C2
12	|12-2| = 10	|12-10| = 2	C2

C1 = {2, 4}

C2 = {10, 12}

Calculat new centroids for formula

$$C = \frac{\text{sum of data points}}{\text{number of data bars}}$$

$$C_1 = \frac{2+4}{2} = 3$$

$$C_2 = \frac{10+12}{2} = 11$$

data points	C1 di	C2 dis	clust
2	|3-2| = 1	|11-2| = 9	C1
4	|3-4| = 1	|11-4| = 7	C1
10	|3-10| = 7	|11-10| = 1	C2
12	|3-12| = 9	|11-12| = 1	C2

C1 = {2, 4}

C2 = {10, 12}

→ K=2 data points 3, 6, 9, 18, 21, 24
initial centroids C1 = 6 C2 = 21

→ : K=2 C1 = 6, C2 = 21

data point	C1 distance	C2 distance	cluster
3	|6-3| = 3	|21-3| = 18	C1
6	|6-6| = 0	|21-6| = 15	C1
9	|6-9| = 3	|21-9| = 12	C1
18	|6-18| = 12	|21-18| = 3	C2
21	|6-21| = 15	|21-21| = 0	C2
24	|6-24| = 18	|21-24| = 3	C2

C1 = {3, 6, 9}

C2 = {18, 21, 24}

$c_2 = \frac{\text{sum of datapoint}}{\text{number of datapoint}}$

$C_1 = \frac{3+6+9}{3} = \frac{18}{3} = 6$

$C_2 = \frac{18+21+24}{3} \quad \frac{63}{3} = 21$

$C_1 = \{3, 6, 9\}$

$C_2 = \{18, 21, 24\}$

datapoint	C_1 distance	C_2 distance	clastor
3	$\|6-3\| = 3$	$\|21-3\| : 18$	C_1
6	$\|6-6\| = 0$	$\|21-6\| = 15$	C_1
9	$\|6-9\| = 3$	$\|21-9\| = 12$	C_1
18	$\|6-18\| = 12$	$\|21-18\| = 3$	C_2
21	$\|6-21\| = 15$	$\|21-21\| = 0$	C_2
24	$\|6-24\| = 18$	$\|21-24\| = 3$	C_2.

$C_1 = \{3, 6, 9\}$

$C_2 = \{18, 21, 24\}$

association

It association rule is an unsupervised learning used to decover relationship b/w data items

ex:- In a supermarket to bye breads may also by butter

terms

1) Support → How frequentaly elements are occuring together
2) confidence → How y is parched x is parched
3) left → It measure how strongly x & y are associated. Compare with the bye y independenty

formul.

Support $(x \to y)$ = Transactions containing both x & y / total Transaction

Confidence $(x \to y)$ = transaction containing both x & y / transaction containing x. $\times 100$

Left $(x \to y)$ = confidence $(x \to y)$ / support (y)

Ex: consider 5 transactions and find out the relationship b/w bread → Butter.

Transaction	item parchaged
T_1	bread → Butter
T_2	bread, butter milk
T_3	bread, milk
T_4	Butter, milk
T_5	Bread, butter

→ Support (Bread → Butter) = $\frac{3}{5} = 0.6$

→ Confidence (Bread → Butter) = $\frac{3}{4} \times 100 = 0.75 \times 100 = 75$

→ support $(y) = \frac{4}{5} = 0.8$

Left (Bread → Butter) = $\frac{0.75}{0.8} = 0.93$

since left is ~~greate than 0~~ less than 1, it is ~~more~~ less likely that when Bread parchaged than Butter will be parchaged

applications

→ e-commorce
→ health care
→ market analesis
→ education

→ consider 6 transaction find the left (M → Bread)

transt.	P
T_1	milk, bread.
T_2	milk, bread, butter
T_3	milk. bread
T_4	milk. butter
T_5	bread, Butter
T_6	milk, bread, butter

→ confidence (milk → bread) = $\frac{4}{6}$ = 0.86 × 100 =

→ lift : (m → b) = $\frac{0.86}{0.83}$ · support (B) = $\frac{5}{6}$ = 0.83

= 0.96

since left is less than 1 ∴ it is less likely.

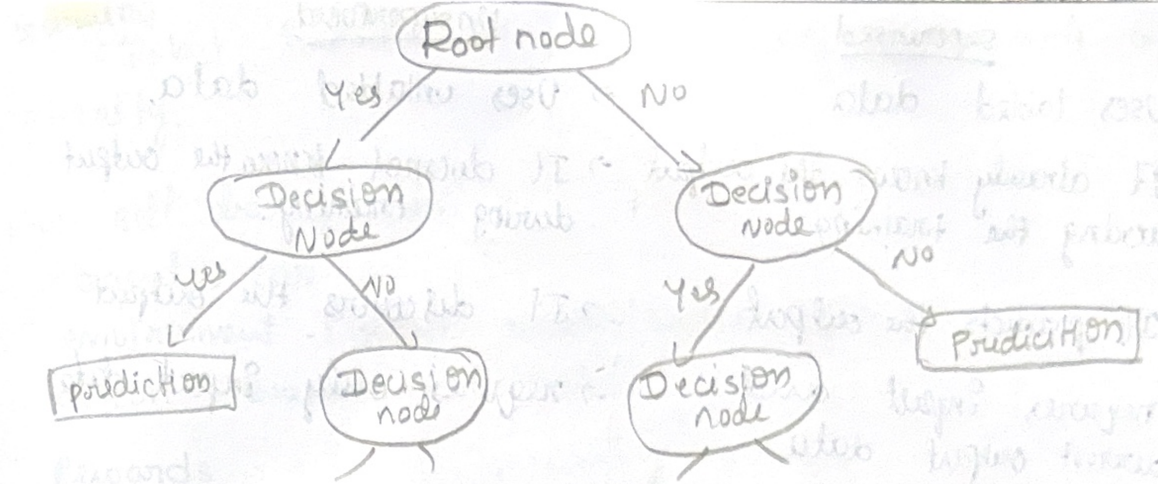
Note:
greater than 1 positive
less than 1 negative
0 : no association

Decision Tree

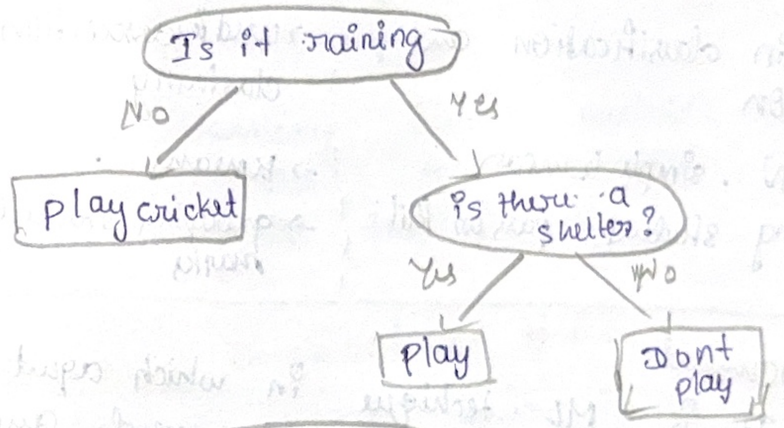
It is a supervised ML technique it uses a tree like structure to make decisions. and predict outcomes based on given condition.

Structure of Decision tree

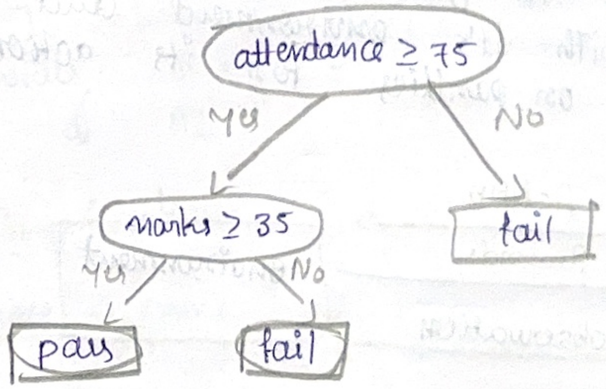
Root node: first and topmost decision in a tree.
Decision node: A node that checks the condition
branch: The path representing the outcome of the condition
Leaf node: final output or prediction.



Ex:-



Ex:-



Ex:- Pass → attendance ≥ 85 marks is = 75
fail. → attendence = 90 marks is = 30
fail → attendence = 60 marks = 90

advantages

→ easy to understand and interpet the data
→ can handle numarical and chatgarical data
→ it requires less data preparation
→ Use full for classification & regration

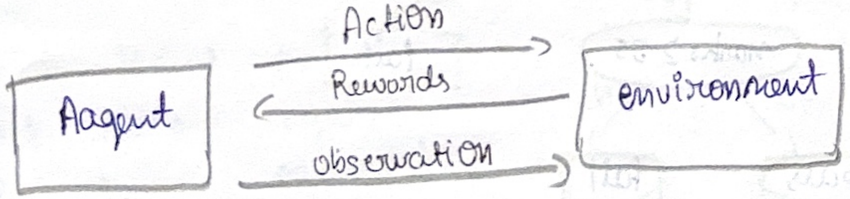
application

→ education
→ healthcare
→ Banking
→ weather forcasting
→ fraud detection
→ e-commarce

Supervised	Unsupervised
→ Uses labled data.	→ Uses unlabled data.
→ It already knows the output during the training	→ It doesnot know the output during training.
→ It predicts the output	→ It discovers the output
→ requires input and correct output data.	→ requires only input data
→ used in classification and regration	→ used for association and clastring
→ Ex:- KNN . simple bankar	→ Kmeans , .
→ predecting student pass or fail.	→ grouping student based on marks

Reinforcement

It is ML techique in which agent learns by interacting with it's environment and reciving rewards or panlties for its action



Aagent

The robot that learns to walk

Environment

the flower of the surrounding

action

Robot Taking step.

Rewards

Robot resives positive reward for walking sccessfuly

panalty

If it falls it get the negative reward

learning

Robot improve it's action base on rewards and panalty.

Ex:- self driving car.

agent -> car

environment -> road, another vachial, peoples.

action -> break , accalaratar.

Rewards -> avoid a predestry safaly.

panalty -> esruns a red signal.

So reinforcement learning learns suitable action throough traile and error to maximise the rewards and minimise the panalty.

Ethical and social implications of AI

It refers to moral issues and effects on individals, society and human life caused by use of AI

Ethical imp lications

-) Transference: => some AI systems work like black box making it default to understand how they reach decisions

-) Security & missuse: AI can be missuse to create fake video and miss information is sprade & carry out cyber attecks

-> Accountability.

It can be default to deside who is responsible when AI system makes harm ful decision

-> privacy.

AI system can collect and analyze large amount of personal data which make affect people privacy.

-> Bias and fareness

AI may produce un fare results if it trained un Baised data.

-> social implication

-> unemployee
-> ~~ifa~~ imfact on education.
-> inequality.
-> human intraction.
-> Improvement in equality of life

Naive Bayes

It is a superwise ML algorithem used for classification it is based on bayes thream ~~as~~ and assume all ~~feature~~ features are independent to each other

Formul:

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

P(A|B) - probality of A when B is know
P(B|A) - probality of B when A is true
P(A) - probality of A
P(B) - probality of B

Ex:

application

-> emile spam detection
-> sentimental analisis
-> fake revise detection
-> document ~~catag~~ catagration
-> news classification

AI Tools

They are software applications that use ~~software integ~~ AI to perform task such as learning, problem solving, decision making, data analysis, Image creation.

Ex:- chat GPT, etc.

Selfdriving car

It is not AI tool it is an application of AI

-> AI tool is software that perform AI related task.
-> AI application is a system that AI uses that slove the real world problem.

AI Libraries

It is a collection of pre written functions and algorithems used to develop AI.

Libraries

-> Tensor flow
-> Pytorch
-> Scikit learn.
-> Numpy
-> Pandas

Tensor flow

-> It is the AI library develop by google.
-> It is used to built and train machine learning and deep learning models.

aplication

->self driving car
->medical analysis
->NLP
->speech regonization
->image regonization

~~Pandas~~ Pandas

-> It is AI library use for data manipulation, data cleaning and data analysis.

uses of pandas

=> It arrganize the data in tables

→ → It handles the missing
→ → Sorting and filter data.

Numpy

It is AI python library. use for numerical calculation and mathamatical operations. and working with multi dimensional arrays

uses

→ creating and manuplating arrays
→ forming mathamatical operations
→ → calculating min, sum, max, avg.

Matklot library.

It is AI python library it is used for representing data. in graphs

uses

→ creating black chat - line chat
→ display data using Histogram
→ visio v.
→ comparing the values.

~~sydic library~~. scikit learn

It is AI python library used for classifical regration, clastering

NLPC Natural language processing)

It is the branch of AI that enables computer to understand interprit process and generate human language such as english, kannado, hindi etc.

Ex:- Google translater, chat bots, online shoping

Text processing

It is the process of converting raw text into clean structured meaningful formate so that computer can analyse and understant it.

Ex: Raw data: The students are playing games!
After text processing: student play game.

steps in text processing

I Tokenization

It divides sentence into smaller unit of words called as tokens

2) Lowercasing

It convert all upper case letters to lowercase letter to maintain consistency.

3) stopword removal

It removes common words that contribute little to a particular text analysis.

Ex: the are

4) Text cleaning

It removes unnecessarey punctuation and special symbols

5) stemming

stemming reduces the words to their root like form by removing prefix and suffix.

Note.
The result may not always be a valid english word

6) Lemmatization

It converts the words into meaningful dictionary base form by considering it meaning and grammatical role.

Ex:- The students are playing games!

1) ["The", "student", "are", "playing", "games", '!']
2) ["the", "student", "are", "playing", "games", "!"]

3) ["student", "playing", "games"]
4) ["student", "playing", "games"]
5) ["student", "play", "game"]
6) ["student", "play", "game"]

applications

→ search engine
→ spam detection
→ document classification
→ machine translation
→ chatbots
→ semantic analysis.

Sentimental analysis.

It is a NLP technique used to identify and classify emotions and opinions express as positive, negative or netural

Types

1) Positive sentiment → express happiness,
2) negative sentiment → anger, disopinment, dissatisfaction
3) nutural sentiment → express facts without opinions +ve or -ve

Ex- +ve = movie is extent
-ve = move is not good
nutural = movie start at 6PM

steps

1) Input text
2) Text processing
3) sentiment classification.
4) output.

Ex:- The product is very good - 1

output :- product very good - 2
+ve sentement. — 3

applications

→ product reviews
→ customer feedback
→ business decision
→ movie reviews
→ social media monitory.

Chatbot

It is a AI based program that interact with user throw text or voice an provides answers using ~~entity~~ NLP.

Type
1) Rule based chatbot → Educational chat bot., banking chatbot
2) AI based chatbot → chatGPS, amazon alex.

Rules based
works on predefined rules and keywords
→ gives fixed responses to specific varsions.

AI based
It uses AI & NLP to understand user queries and generate appropriate response.
→ Can handle different type of asking question.

applications
→ customer support
→ banking services
→ travel booking & reservations

Robotics

It is the branch of technology that deals with designing, constracting, programing and operation of the Roborts to perform fast automaticaly with minimum assistance of human.

Ex:- industrial roborts -> use in factory -> valding, painting, packaging
medical roborts -> in which assist doctors during surgery.
domastic roborts -> vacum cleaner, it perform house hold task such as floor cleaning
human roborts -> design to perform task and interact with human like ways
space roborts -> It explores planets collects sentific data and perform task in space

Main components of robots

1) sensors
2) actuactors
3) power supply
4) controllers
5) end effectors

applications

1 -> industries
-> medical
0 -> house
1 -> agriculture.
E. -> space

Computer vision

It is the branch of AI the enables computer to identify, analysis and understanding the video and images.

Ex:- facial regognization ->
Self driving car
medical image analysis.
object recognization.

steps

1) image input :- A camara capcutes a image or video.
2) image pre processing :- The image is prepared for analysis.
3) feature extraction & analysis :- the system analysis shapes, pattens, color. or learned feature.
4) recognization :

applications

-> health care
-> autonomes vehical
-> security
-> detail & product regognization.